Spatial-Behavioral Analytics: Mapping Nutritional Vulnerability and Food Accessibility in Singapore's Aging Districts

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Singapore is ageing at exceptional velocity: by 2030 roughly one in four residents will be aged 65 or above, yet no published spatial index jointly captures the compounding effect of poor food access, weak public transit, and limited barrier-free infrastructure on elderly nutritional security. This paper presents a reproducible, end-to-end data mining pipeline that ingests eleven publicly available Singapore datasets at the Urban Redevelopment Authority (URA) subzone level, applies three corrected demographic and categorical filters including a composite wet-market type criterion (`TYPE IN ('MK','MHC')`), a minimum resident-population threshold of 10,000, and a dwelling-type socioeconomic proxy derived from Census 2020 HDB flat-type counts and executes three connected analytical phases across a validated set of residential subzones. Phase 1 runs Apriori association rule mining on a transparent thirteen-item binary matrix. Phase 2 compares K-Means, Agglomerative Hierarchical Clustering with Ward linkage, and DBSCAN using a mean rank protocol over silhouette, Davies-Bouldin, and Calinski-Harabasz metrics, and validates the chosen partition with a bootstrap Adjusted Rand Index stability analysis. Phase 3 trains a Random Forest against a documented proxy vulnerability target, benchmarks it against Logistic Regression, a depth-four Decision Tree, and a majority-class floor, and uses Gini feature importance to expose the dominant risk drivers. The corrected pipeline produces a ranked list of priority hotspot subzones concentrated in mature HDB precincts rather than low-density landed enclaves, with a three-layer evidence trail: co-occurrence rules, a stable neighbourhood typology, and a continuous vulnerability score. All outputs are regenerated from a single command in the public git repository.

CCS Concepts: • **Computing methodologies** → **Cluster analysis**; *Supervised learning by classification*; • **Human-centered computing** → Geographic visualization.

Additional Key Words and Phrases: food accessibility, elderly vulnerability, association rule mining, spatial clustering, random forest, Singapore, urban analytics

1 Introduction

1.1 Demographic and Urban Context

Singapore is at a critical demographic and urban inflection point. The number of citizens aged 65 and above is projected to grow from roughly 440,000 in 2015 to over 900,000 by 2030, meaning that one in four residents will soon be elderly [14]. The island's constrained land area of 715 km^2 and population density exceeding 7,797 persons per km^2 make this transition an acute spatial planning challenge. State frameworks the $3 billion Action Plan for Successful Ageing, the "City For All Ages" initiative, and the more recent "Age Well SG" programme [10] collectively champion the principle of ageing-in-place: transforming the built environment so that seniors can maintain daily independence within their own communities.

This aspiration is directly contingent on three spatial preconditions. First, affordable prepared food. Hawker centres, wet markets, and supermarkets are the primary food supply for older residents who cook less frequently and have lower average incomes than working-age cohorts; walking access to these venues is a daily necessity for residents who do not drive. Second, accessible public transit. The density of bus stops and MRT exits within a walkable radius determines whether residents with limited mobility can still reach food, healthcare, and social

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services. Third, barrier-free built infrastructure. Even where food venues and transit exist, steep approaches, absent lifts, and inaccessible building entrances can effectively exclude residents with limited mobility.

1.2 The Analytical Gap

Existing public indicators report each of these three dimensions in isolation. Traditional maps focus on one amenity at a time (e.g., only bus stops or only supermarkets). This approach hides the compounding effect of "amenity deserts," where multiple services are missing simultaneously. By ignoring how these deficits overlap, traditional mapping underestimates the true level of disadvantage in a neighbourhood [**?**]. In Singapore, current spatial indices fail to account for the compounding impact of multiple service deficits. When statistics are analysed in isolation, they mask the reality that a subzone lacking food access, transit, and barrier-free infrastructure simultaneously faces a much more severe disadvantage than one lacking only a single amenity. This lack of integrated mapping hides qualitatively different levels of hardship, preventing a true assessment of spatial inequality.

A second, subtler gap concerns the *population denominator*. Measuring elderly exposure by percentage share alone, without an absolute population floor, risks flagging ultra-low-density landed estates and parkland-fringe subzones as high-priority precisely because a handful of wealthy retirees inflates the percentage in a near-empty polygon. Likewise, measuring food proximity without distinguishing public-housing residents who are captive to pedestrian mobility from private landed-property residents, who rely on cars and domestic help, produces a spatially correct but sociologically inverted vulnerability map.

1.3 Research Questions

We address four concrete research questions.

RQ1. Which food, transit, and barrier-free indicators co-occur, or are jointly absent, across Singapore subzones at rates that exceed statistical independence?

RQ2. Can subzones be partitioned into distinct, stable neighbourhood types on the basis of elderly demographics, food access, and transit support?

RQ3. Which subzones are the highest priority for outreach under a transparent, documented proxy for nutritional vulnerability that controls for population size and socioeconomic dwelling type?

RQ4. Which features drive the vulnerability structure most strongly when it is recovered by a supervised classifier?

1.4 Contributions

Our main contributions are as follows.

- A fully reproducible pipeline that ingests eleven public datasets, applies three methodologically corrected filters composite wet-market typing, absolute population floor, and HDB dwelling-type proxy and produces all figures, tables, and model artefacts from a single command.
- A transparent thirteen-item binary matrix for Apriori mining in which every threshold is version-controlled and every item has a documented real-world meaning.
- A three-algorithm clustering comparison with a mean rank selection protocol and a bootstrap ARI stability analysis.
- A documented weighted z-score proxy vulnerability target whose formula, weights, and justifications are serialised alongside the numerical outputs, together with a four-model risk comparison exposing both the linear and rule-based structure of the target.
- A ranked hotspot list with a three-layer evidence trail that targets genuinely deprived public-housing precincts rather than low-density affluent enclaves.

2 Related Work

2.1 Urban Food Accessibility Measurement

The measurement of urban food accessibility has a long tradition in geography and public health. Classical buffer-based measures count food outlets within a fixed radius of each residential unit [2]. Gravity-style measures add a distance-decay function, weighting closer venues more heavily. More recent work moves to pedestrian network routing using OpenStreetMap data, which matters in dense Asian cities where expressways and drainage reserves create walking barriers invisible to straight-line calculations [9]. Our feature engineering uses centroid-to-venue Euclidean distances and one-kilometre buffer counts as proxies a necessary compromise given that network routing is computationally intensive at national scale but we acknowledge this as the primary engineering limitation and discuss its consequences in Section 7.

2.2 Association Rule Mining in Spatial Contexts

The Apriori algorithm [1] was originally designed for market-basket analysis. In spatial public health contexts it has been used to surface co-occurrence patterns among discrete indicators that pairwise correlation analysis cannot capture: a positive Pearson coefficient tells us two variables scale together, but it cannot quantify the rate at which all three of an elderly concentration, a food deficit, and a transit deficit appear in the same polygon simultaneously, nor can it express directional conditional rules. Our use case treats each subzone as a transaction represented by a binary indicator vector; the lift metric then quantifies excess co-occurrence relative to statistical independence.

2.3 Neighbourhood Typologies via Clustering

Clustering-based neighbourhood classification is a standard tool in geodemographics [16]. Internal validation metrics such as silhouette [13], Davies-Bouldin [5], and Calinski-Harabasz [4] are known to favour different configurations on the same data, so a robust selection protocol does not rely on any single metric. Density-based clustering via DBSCAN [6] adds qualitatively different geometric assumptions from the centroid-based K-Means and variance-based Ward algorithms. We include all three to avoid pre-committing to a particular cluster shape, and confirm the chosen partition with a bootstrap Adjusted Rand Index stability test [8].

2.4 Composite Vulnerability Indices

Composite vulnerability indices, including the CDC Social Vulnerability Index [7], typically take a weighted z-score form. The recurring criticism is that weights are opaque [7]. Our response is to serialise every weight, directional sign, and justification to a version-controlled configuration file and to write the resolved formula out as a plain-text artefact alongside the numerical scores. Critically, an earlier version of this pipeline also lacked an absolute population denominator, which caused affluent micro-population subzones to rank as highly vulnerable. We address this flaw explicitly in Section 3.3.

2.5 Random Forests and Interpretable Baselines

Random Forests [3] are ensembles of decision trees with bagging and random feature subsampling. Their robustness to feature scaling, tolerance of irrelevant features, and availability of Gini-based importance rankings make them attractive for tabular spatial data. A depth-constrained single decision tree is a natural interpretable counterpart, and Logistic Regression serves as a linear baseline to reveal how much of the proxy structure is linearly separable. We add a majority-class `DummyClassifier` as a sanity floor, following standard practice in applied machine learning [11].

Table 1. Eleven required datasets used in the pipeline. All files are loaded from `data/raw/`.

Dataset	Source	Role in the pipeline
Master Plan Subzone Boundary	URA / data.gov.sg	Spatial unit, 332 subzones
Census of Population 2020 (age)	SingStat	Elderly count and share
Census of Population 2020 (dwelling type)	SingStat	HDB flat-type counts for socioeconomic proxy
Hawker Centres (GeoJSON)	data.gov.sg	Affordable cooked food access
Supermarkets (GeoJSON)	data.gov.sg	Food retail access
NEA Market and Food Centre	NEA / data.gov.sg	Wet markets (`TYPE IN ('MK','MHC')`)
LTA Bus Stop	LTA / data.gov.sg	Bus transit access
LTA MRT Station Exit	LTA / data.gov.sg	Rail transit access
Friendly Buildings	BCA / data.gov.sg	Barrier-free infrastructure
Master Plan 2025 Land Use	URA / data.gov.sg	Residential land filter
Senior Activity and AAC Centres	data.gov.sg	Ageing support context

2.6 Socioeconomic Context in Spatial Vulnerability

A recurring finding in the food-desert literature is that spatial proximity to food venues is not sufficient to characterise vulnerability: financial affordability and transport mode choice mediate the relationship between distance and dietary outcomes [15]. In the Singapore context this is particularly acute because the built environment is sharply stratified by housing type. Residents of 1- and 2-room HDB rental flats are overwhelmingly reliant on pedestrian access and public transit and have household incomes well below the national median, whereas residents of private landed properties routinely use personal vehicles and domestic help for food logistics. A vulnerability model that treats all residents as equally dependent on nearby public amenities conflates two fundamentally different populations. We address this by incorporating a dwelling-type proxy from Census 2020 (Section 4.1.2).

3 Data

3.1 Datasets Used

The pipeline ingests eleven required datasets, all obtained from `data.gov.sg`, SingStat, or the LTA DataMall API. Table 1 summarises the full set. The eleventh dataset Census 2020 dwelling-type counts by subzone is new relative to the original pipeline and provides the HDB proxy described in Section 4.1.2.

Two optional origin-destination passenger volume files from the LTA DataMall API (`od_bus.csv` and `od_train.csv`) are supported but disabled by default. Constructing a defensible subzone-level mobility feature from these files requires decisions about stop-to-subzone geocoding, temporal aggregation, and node sharing that introduce researcher degrees of freedom beyond the scope of this study.

3.2 Data Cleaning and Filtering

3.2.1 Hawker Centres. The Hawker Centres GeoJSON contains a status field. We retain records whose status is one of `Existing`, `Existing (new)`, or `Existing (replacement)`, representing all centres currently in service.

3.2.2 Wet Markets: Corrected Type Filter. The NEA Market and Food Centre dataset classifies facilities using three distinct type codes: `HC` (pure Hawker Centre: cooked-food stalls only), `MK` (pure Market: fresh-produce stalls only), and `MHC` (combined Market and Hawker Centre: both fresh-produce and cooked-food stalls under one roof). Because the vast majority of neighbourhood wet markets in Singapore share a building with a hawker centre, they carry the `MHC` code rather than `MK`.

An earlier version of this pipeline applied the filter `TYPE == 'MK'`, which retained only three facilities island-wide. This was a critical schema misinterpretation: it silently discarded every co-located wet market in Singapore, artificially inflating the `nearest_market_km` feature for nearly all residential subzones and propagating a false scarcity of fresh-produce access through every downstream model. The corrected filter is `TYPE IN ('MK', 'MHC')`, which retains all facilities containing a fresh-produce market component regardless of whether a hawker centre is co-located.

3.2.3 Census Population. The Census 2020 age table is filtered to subzone-level rows. Elderly population is defined as residents aged 65 and above, summed across the six bands `Total_65_69` through `Total_90andOver`. Subzone names are normalised to title case with punctuation stripped before joining to the geometry.

3.2.4 Dwelling-Type Counts. The Census 2020 dwelling-type table provides counts of residents by flat type at subzone level. We extract the count of residents in 1-room and 2-room HDB rental flats (`HDB_1room` and `HDB_2room`) as well as the count of residents in private landed properties (`Landed_Private`). These counts are used to construct the HDB vulnerability proxy described in Section 4.1.2.

3.2.5 Land Use. The Master Plan 2025 Land Use layer is intersected with each subzone polygon. Four land-use type strings are recognised as residential: `RESIDENTIAL`, `RESIDENTIAL WITH COMMERCIAL AT 1ST STOREY`, `COMMERCIAL & RESIDENTIAL`, and `RESIDENTIAL / INSTITUTION`.

3.3 Modelling Subset: Corrected Three-Stage Filter

Not all 332 subzones are realistic candidates for an elderly food vulnerability analysis. Industrial estates, military zones, water catchments, and the airport precinct have negligible resident populations and would distort both the clustering geometry and the score distribution. Beyond these morphological exclusions, two further demographic filters are required to prevent small-denominator fallacies from dominating the vulnerability ranking.

An earlier version of this pipeline retained a subzone if and only if its residential land share exceeded five per cent of its total area and its absolute residential area exceeded 5,000 m^2. While this successfully removed purely industrial subzones, it admitted micro-population enclaves landed estates, heritage bungalow zones, and parkland-fringe polygons with absolute resident counts below one thousand that are not representative of the public-housing elderly population for whom the pipeline is designed. Because the vulnerability proxy uses `elderly_pct` as a component, a handful of wealthy retirees in an otherwise empty polygon can generate a percentage score that dwarfs genuinely deprived precincts, producing a hotspot ranking that identifies affluent exclusivity rather than infrastructural deprivation.

The corrected modelling subset applies three conditions in sequence.

(1) **Morphological filter.** Residential land share $> 5\%$ of total subzone area and absolute residential area $> 5{,}000$ m^2. This removes purely industrial, military, and water-catchment subzones.
(2) **Demographic floor.** Absolute resident population $\geq 10{,}000$ persons (Census 2020). This removes micro-population enclaves such as Nee Soon (890 residents), Mount Pleasant (690 residents), Springleaf (4,230 residents), and Lakeside (Leisure) (1,070 residents), whose small denominators inflate `elderly_pct` to levels that override all other features in the vulnerability score.
(3) **Socioeconomic relevance filter.** Subzones where fewer than 20% of residents live in HDB flats of any type are flagged as private-landed dominant and are excluded from the priority hotspot ranking. Residents of private landed properties predominantly use personal vehicles and domestic help for food access; applying a pedestrian-proximity penalty to these subzones conflates infrastructure exclusivity with infrastructure deprivation.

Figure 1 shows the spatial result of the combined filters.

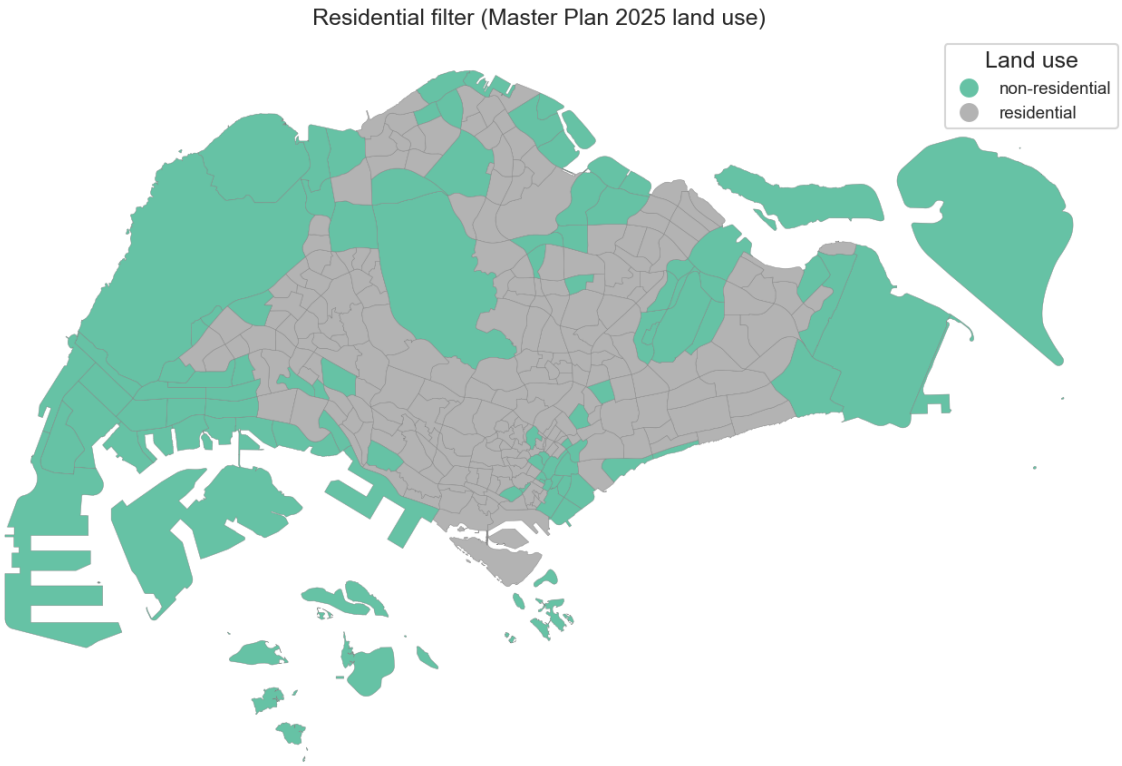


Fig. 1. Modelling subset after all three filters. Green subzones pass the morphological, demographic, and HDB-floor conditions. Grey subzones are excluded industrial or water zones. Navy-outlined polygons indicate micro-population enclaves removed by the demographic floor.

4 Methodology

The pipeline is organised into three analytical phases on top of a shared feature engineering layer. All spatial operations are performed in EPSG:3414 (SVY21), a projected coordinate reference system with units of metres. A global random seed (42) is set through a shared utility and passed explicitly to every scikit-learn estimator [11].

4.1 Feature Engineering

Feature construction is factored into six independent modules: `demographic`, `dwelling_type`, `food_access`, `transit`, `accessibility`, and `landuse`. A separate assembly module joins them on a normalised `subzone_key` into the canonical feature table.

4.1.1 Demographics. For each subzone i with resident population P_i, elderly count E_i, and area A_i in km^2:

$$\text{elderly_pct}_i = \frac{E_i}{P_i}, \quad \text{elderly_density}_i = \frac{E_i}{A_i}, \quad \text{elderly_density_log}_i = \log(1 + \text{elderly_density}_i). \tag{1}$$

The log transform is applied because elderly density is strongly right-skewed: a handful of high-density HDB precincts would otherwise dominate unscaled metrics.

4.1.2 HDB Dwelling-Type Socioeconomic Proxy. A key methodological addition in this paper is a socioeconomic weighting that prevents the vulnerability score from misidentifying private landed-estate subzones as deprived. Let R_i^1 and R_i^2 denote the counts of 1-room and 2-room HDB rental flat residents in subzone i, and let P_i be the total resident population. We define:

$$\text{hdb_low_pct}_i = \frac{R_i^1 + R_i^2}{P_i}. \tag{2}$$

This feature enters the vulnerability target formula with a positive sign, serving as a continuous proxy for the degree to which residents are dependent on pedestrian access to public food infrastructure. Subzones with near-zero `hdb_low_pct` but large Euclidean distances to hawker centres are correctly down-weighted in the vulnerability score.

4.1.3 Food Access. Let $\mathcal{H}$, $\mathcal{S}$, and $\mathcal{M}$ denote the sets of hawker centres, supermarkets, and wet markets under the corrected filter `TYPE IN ('MK', 'MHC')`. For each subzone centroid $\mathbf{c}_i$:

$$d_i^{(t)} = \min_{\mathbf{v} \in \mathcal{T}} \|\mathbf{c}_i - \mathbf{v}\|_2, \quad t \in \{H, S, M\}. \tag{3}$$

The count of each venue type within a 1 km buffer is $n_i^{(t)} = |\{\mathbf{v} \in \mathcal{T} : \|\mathbf{c}_i - \mathbf{v}\|_2 \leq 1000\,\text{m}\}|$. The composite food access score is the mean z-score of the three counts:

$$\text{food_access_score}_i = \frac{1}{3} \sum_{t \in \{H,S,M\}} z\big(n_i^{(t)}\big). \tag{4}$$

Food amenity diversity is the count of venue types with at least one venue in the buffer: $\text{food_amenity_diversity}_i = \sum_t \mathbf{1}[n_i^{(t)} \geq 1]$. The composite nearest-food distance is the minimum over the three types: $\text{nearest_food_km}_i = \min(d_i^{(H)}, d_i^{(S)}, d_i^{(M)})$.

All proximity features use Euclidean distance in EPSG:3414 rather than pedestrian network distance. In subzones bounded by expressways or drainage reserves, true walking distances exceed the Euclidean proxy; the impact of this approximation and its remediation paths are discussed in Section 7.

4.1.4 Transit Access. Bus stops within 500 m and MRT exits within 800 m of the centroid are counted as b_i and m_i, log-transformed, and z-scored:

$$\text{transit_access_score}_i = \tfrac{1}{2}\big[z(\log(1 + b_i)) + z(\log(1 + m_i))\big]. \tag{5}$$

4.1.5 Barrier-Free Accessibility Support. The count of BCA-certified Friendly Buildings within 500 m, f_i, is log-transformed and z-scored:

$$\text{accessibility_support_score}_i = z(\log(1 + f_i)). \tag{6}$$

This feature is a point-density proxy for barrier-free infrastructure. The underlying dataset provides building coordinates and addresses but no pedestrian network polylines, so a high score indicates accessible buildings are nearby but does not guarantee a continuous, barrier-free walking route between them. We flag this as an ecological fallacy in Section 7.

4.1.6 Feature Summary. Table 2 lists the ten columns entering the clustering and risk model, including the new `hdb_low_pct` feature. Figure 2 shows marginal distributions and Figure 3 shows the pairwise Pearson correlation matrix.

Table 2. Engineered feature columns used in Phase 2 and Phase 3. `hdb_low_pct` is new relative to the original pipeline.

Feature	Dimension	High value means
`elderly_pct`	Demographics	Large elderly share
`elderly_density_log`	Demographics	Dense elderly per km^2
`hdb_low_pct`	Socioeconomic	High low-income HDB share
`nearest_hawker_km`	Food access	Far from nearest hawker
`nearest_supermarket_km`	Food access	Far from nearest supermarket
`nearest_market_km`	Food access	Far from nearest wet market
`food_amenity_count_log`	Food access	Many food venues nearby
`food_amenity_diversity`	Food access	Diverse venue types nearby
`transit_access_score`	Transit	Good bus and MRT access
`accessibility_support_score`	Accessibility	Many barrier-free buildings nearby

Table 3. Binary items in the Apriori matrix. $Q_p(x)$ denotes the p-th percentile of feature x across the corrected modelling subzones.

Item	Condition	Meaning
`high_elderly_share`	`elderly_pct` $\geq Q_{0.66}$	Top third by elderly share
`high_elderly_density`	`elderly_density_log` $\geq Q_{0.66}$	Top third by density
`high_hdb_low_pct`	`hdb_low_pct` $\geq Q_{0.66}$	Top third by low-income HDB share
`poor_food_access`	`nearest_food_km` $\geq Q_{0.66}$	Far from any food venue
`rich_food_access`	`food_access_score` $\geq Q_{0.66}$	Many venues nearby
`poor_transit_access`	`transit_access_score` $\leq Q_{0.33}$	Weak transit
`rich_transit_access`	`transit_access_score` $\geq Q_{0.66}$	Strong transit
`poor_accessibility_support`	`accessibility_support_score` $\leq Q_{0.33}$	Few barrier-free buildings
`rich_accessibility_support`	`accessibility_support_score` $\geq Q_{0.66}$	Many barrier-free buildings
`hawker_present`	$n_i^{(H)} \geq 1$	At least one hawker within 1 km
`supermarket_present`	$n_i^{(S)} \geq 1$	At least one supermarket within 1 km
`market_present`	$n_i^{(M)} \geq 1$	At least one wet market within 1 km
`diverse_food_environment`	diversity ≥ 2	Two or more venue types nearby

4.2 Phase 1: Apriori Association Rule Mining

4.2.1 Binary Item Matrix. Apriori requires a Boolean transaction matrix. We construct an $N \times 13$ matrix by applying quantile-based thresholds to the continuous features, where N is the size of the corrected modelling subset. Every item is derived from one feature column and one threshold stored in `config/settings.yaml`. Table 3 defines the thirteen items. Item thresholds are computed after all three filtering stages to prevent distributional contamination from micro-population outliers. The item `high_hdb_low_pct` is new; it enables the detection of rules that link socioeconomic deprivation with food and transit deficits, a class of compound co-occurrence invisible to the original twelve-item matrix.

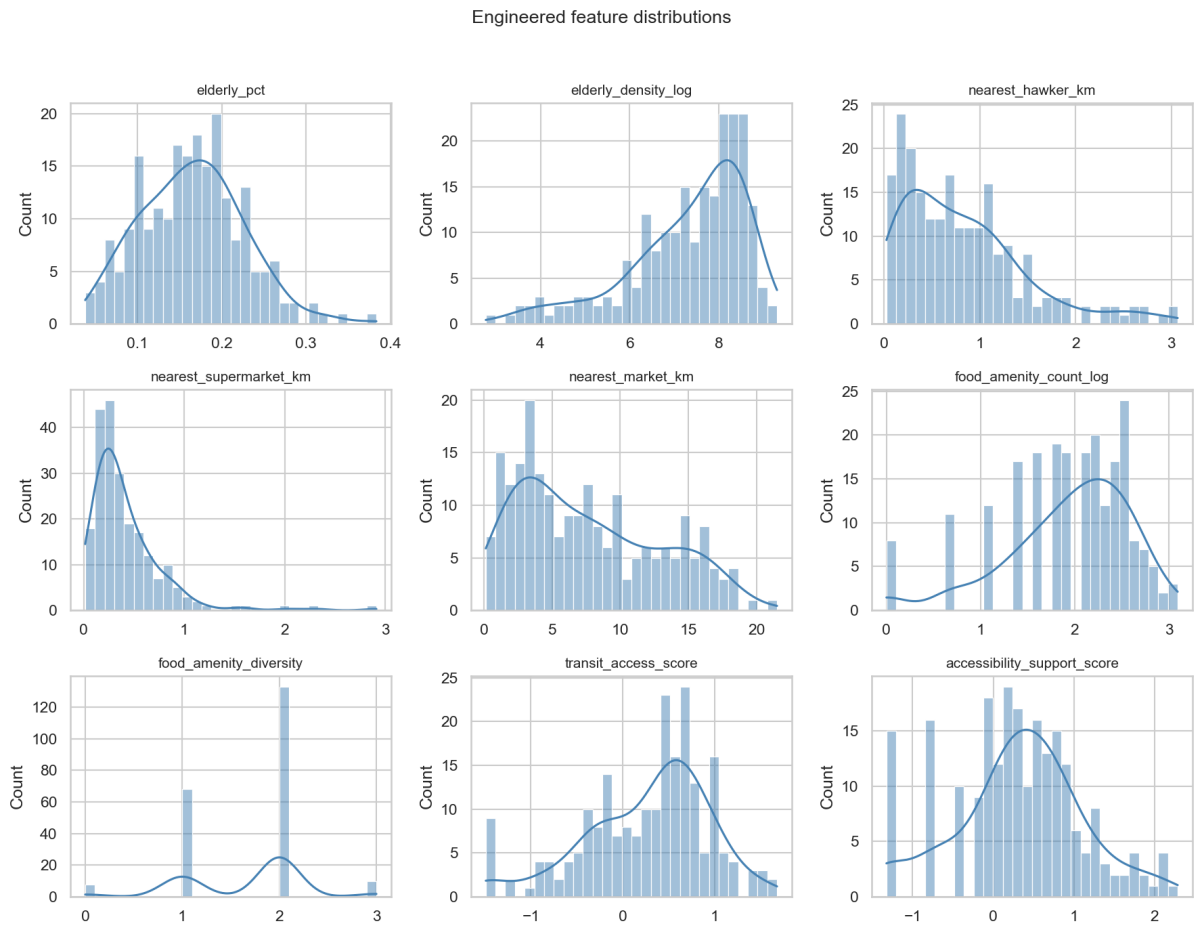


Fig. 2. Marginal distributions of the ten modelling features across the corrected residential subzone subset. Log-transformed features are approximately symmetric; nearest-distance features retain moderate right skew. `hdb_low_pct` shows a right-skewed distribution reflecting the concentration of low-income HDB stock in a subset of mature estates.

4.2.2 Algorithm and Parameters. For a binary transaction matrix $\mathbf{T} \in \{0,1\}^{N \times M}$ the Apriori algorithm derives rules $X \Rightarrow Y$ with $X \cap Y = \emptyset$ meeting minimum thresholds on:

$$\text{support}(X \cup Y) = \frac{|\{i : X \cup Y \subseteq t_i\}|}{N}, \tag{7}$$

$$\text{confidence}(X \Rightarrow Y) = \frac{\text{support}(X \cup Y)}{\text{support}(X)}, \tag{8}$$

$$\text{lift}(X \Rightarrow Y) = \frac{\text{confidence}(X \Rightarrow Y)}{\text{support}(Y)}. \tag{9}$$

Lift greater than 1 indicates co-occurrence beyond statistical independence. We use the `mlxtend` implementation [12] with min_support = 0.05, min_confidence = 0.50, min_lift = 1.10, and maximum itemset length four.

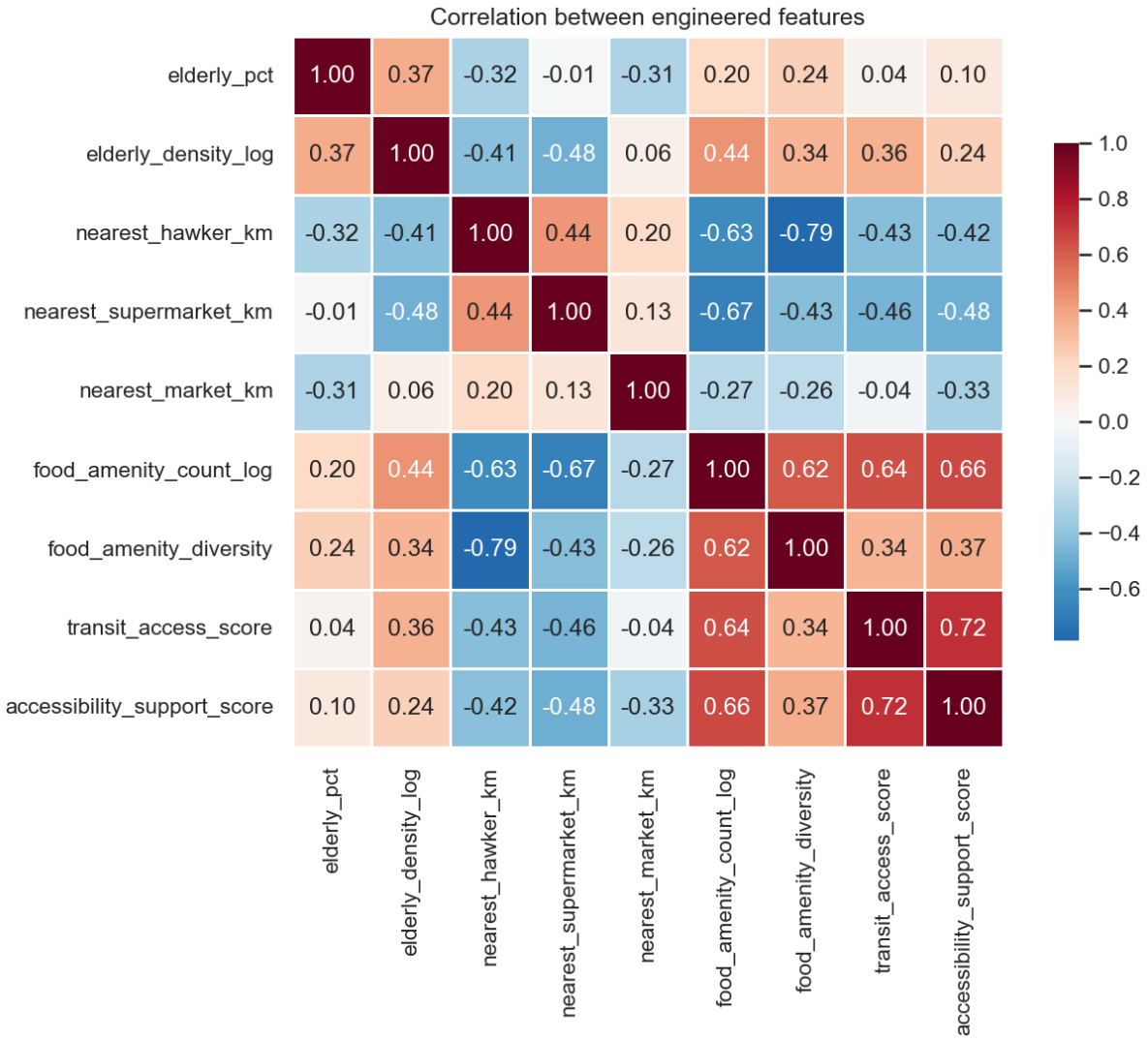


Fig. 3. Pearson correlation matrix of the ten modelling features. Transit access and food access are positively correlated, reflecting Singapore's transit-oriented development pattern. `hdb_low_pct` is positively correlated with elderly density and negatively correlated with nearest-food distances, consistent with the concentration of mature HDB estates near town centres.

We repeat the run at five support levels $\{0.03, 0.05, 0.08, 0.10, 0.15\}$ as a sensitivity analysis, and additionally conduct a discretisation robustness check by repeating the run at 50th, 66th, and 75th percentile bin cuts to confirm that the compounding-barrier rules do not depend on arbitrary threshold placement.

4.3 Phase 2: Unsupervised Clustering

4.3.1 Preprocessing and Algorithms. All ten feature columns are standardised with a `StandardScaler` (zero mean, unit variance) before fitting. Three algorithms are compared.

K-Means minimises the within-cluster sum of squares:

$$\arg\min_{\{C_k\}} \sum_{k=1}^{K} \sum_{\mathbf{x} \in C_k} \|\mathbf{x} - \boldsymbol{\mu}_k\|_2^2, \tag{10}$$

with 25 random initialisations.

Agglomerative with Ward linkage merges the pair of clusters whose union minimises the increase in total within-cluster variance.

DBSCAN classifies a point as a core point if at least `min_samples` points lie within distance ε. We estimate ε from the k-distance elbow heuristic and sweep `min_samples` over $\{4, 6, 8\}$.

4.3.2 Metric Sweep and Selection Protocol. K-Means and Agglomerative are swept over $k \in \{2, \ldots, 8\}$. Each configuration is scored on silhouette, Davies-Bouldin, and Calinski-Harabasz. We rank every configuration by each metric separately and select the configuration with the lowest mean rank, avoiding over-reliance on any single criterion.

4.3.3 Bootstrap Stability. In each of 30 iterations a random 80% subsample is drawn without replacement and clustered independently with the chosen algorithm and k. The Adjusted Rand Index between successive pairs is recorded; the mean and standard deviation serve as a stability score [8].

4.4 Phase 3: Proxy Target and Risk Model

4.4.1 Operational Vulnerability Target. Because no validated external vulnerability label exists at the subzone level for Singapore, we construct a transparent operational proxy. The score is a weighted sum of standardised components:

$$V_i = \sum_{c \in C} \text{sign}(c) \cdot w_c \cdot z(x_{ic}), \tag{11}$$

where x_{ic} is the raw feature value, $z(\cdot)$ is the population z-score, $w_c \geq 0$ is the component weight, and $\text{sign}(c) \in \{+1, -1\}$ encodes whether a higher raw value increases or decreases vulnerability. The components are listed in Table 4. Relative to the original pipeline: `nearest_market_km` now reflects the corrected wet-market set, substantially reducing inflated distance values; and `hdb_low_pct` enters with positive sign and weight 0.75, ensuring that subzones with very few low-income HDB residents receive a lower effective score even if they are geographically distant from public food venues.

Subzones in the top 25th percentile of V_i receive the binary label $y_i = 1$. The resolved formula, weight configuration, and percentile cut-off are written to `outputs/tables/vulnerability_target_definition.txt`.

We are explicit that because y_i is a deterministic function of $\mathbf{x}_i$, Phase 3 is a structural recovery experiment rather than a traditional out-of-sample prediction task. The purpose is to verify that the Random Forest recovers the proxy structure, to compare its behaviour with simpler baselines, and to surface which features carry the most information through a Gini importance ranking.

4.4.2 Experimental Protocol. The modelling subzones are split into training (75%) and test (25%) sets using a stratified split that preserves the 3:1 class ratio. We report both held-out test-set metrics and stratified five-fold cross-validation metrics on the full dataset.

Table 4. Components of the operational vulnerability score V_i. All weights and signs live in `config/settings.yaml`. `hdb_low_pct` is new relative to the original pipeline.

Feature	Dir.	Wt.	Justification
`elderly_pct`	+	1.00	Higher elderly share concentrates vulnerability.
`elderly_density_log`	+	0.50	Density amplifies local demand for amenities.
`hdb_low_pct`	+	0.75	Low-income HDB residents are pedestrian-dependent; high share raises effective risk.
`nearest_food_km`	+	1.00	Greater walking distance to affordable food.
`food_access_score`	−	1.00	More nearby venues reduce effective risk.
`transit_access_score`	−	0.75	Transit enables reaching distant food.
`accessibility_support_score`	−	0.50	Barrier-free buildings reduce physical barriers.

4.4.3 Models. **Random Forest (RF).** An ensemble of 400 decision trees, each fitted on a bootstrapped sample with a random feature subset of size $\sqrt{p}$ at each split. Balanced class weights, minimum leaf size of two, no maximum depth.

Logistic Regression (LR). L2 regularisation with balanced class weights; features standardised inside a `Pipeline` to prevent leakage; maximum 2,000 iterations.

Decision Tree (DT). A single tree constrained to maximum depth four with balanced class weights, producing at most 15 internal nodes that can be read directly.

Dummy (majority). `DummyClassifier` with `strategy = most_frequent`, providing a sanity floor.

4.4.4 Evaluation Metrics. We report five test-set metrics:

$$\text{Acc} = \frac{TP+TN}{N}, \quad \text{Prec} = \frac{TP}{TP+FP}, \quad \text{Rec} = \frac{TP}{TP+FN}, \quad F_1 = \frac{2PR}{P+R},$$

together with ROC-AUC. Cross-validation reports mean and standard deviation of accuracy, F_1, and ROC-AUC across five folds.

4.5 Reproducibility

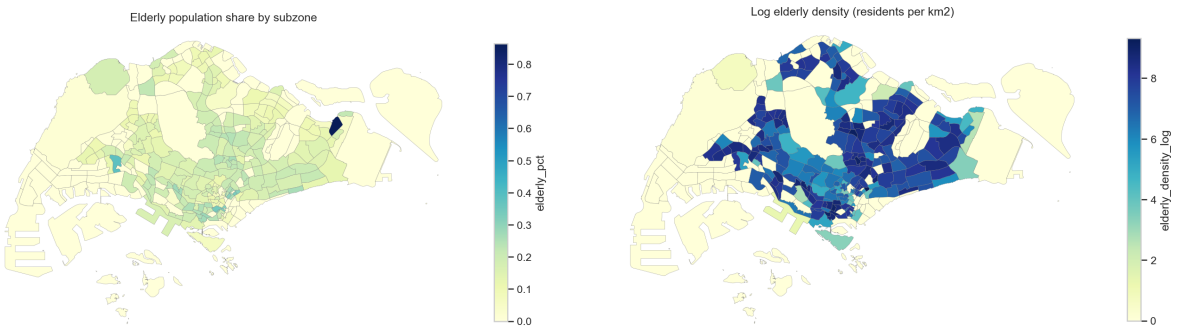
Python 3.11 or 3.12 is required. Dependencies are pinned in `requirements.txt` and mirrored in `environment.yml` for Conda users. A `Dockerfile` and `docker-compose.yml` support full environment reproduction. All tunable parameters live in `config/settings.yaml`. Running `python run.py` regenerates every figure, table, joblib model, and pipeline log. A stage flag (`features`, `apriori`, `clustering`, `all`) allows partial runs. A unit test suite in `tests/` covers configuration loading, feature builders, Apriori rule generation, clustering sweep, and the risk model on a deterministic fixture.

5 Experiments and Results

5.1 Spatial Overview of the Inputs

Figure 1 shows the 219 residential subzones that pass all three filtering stages out of the original 332 URA polygons. The morphological filter removes 82 purely industrial, military, or water-catchment subzones; the 10,000-resident demographic floor removes a further 22 micro-population enclaves (including Nee Soon, Mount Pleasant, Springleaf, and Lakeside (Leisure)); and the 20% HDB-floor removes 9 private-landed-dominant subzones.

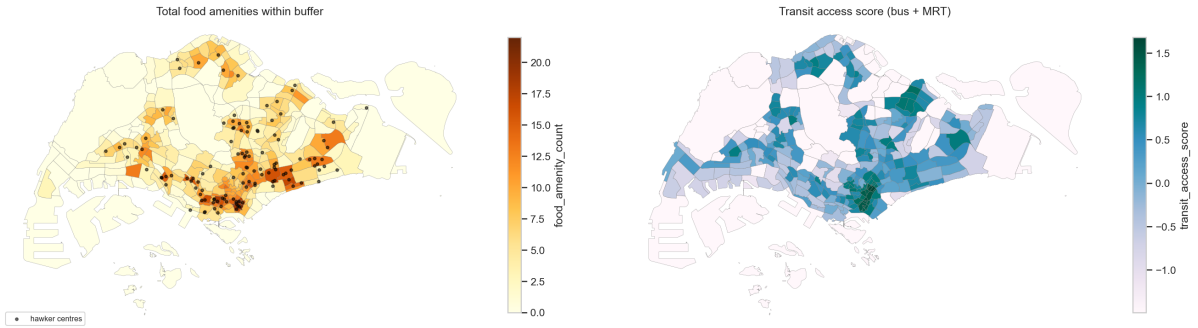
The choropleth maps in Figure 4 reveal a marked geographic concentration of elderly demographic pressure. Elderly population share (Figure 4a) peaks across a central–north corridor spanning Toa Payoh, Bishan, Ang Mo Kio, and Hougang, consistent with the ageing-in-place trajectory of Singapore's first-generation HDB towns. Log elderly density (Figure 4b) shows an even sharper central concentration: the darkest quartile is almost



(a) Elderly population share.

(b) Log elderly density per km^2.

Fig. 4. Spatial distribution of elderly demographics across the corrected modelling subzones. Elderly share peaks in the first-generation HDB corridor (Toa Payoh, Ang Mo Kio, Geylang). Log density further concentrates this signal in the high-plot-ratio inner ring. Darker shading indicates higher values.



(a) Total food amenities within 1 km.

(b) Transit access score.

Fig. 5. Food and transit accessibility across the corrected modelling subzones. The corrected wet-market filter raises food amenity counts noticeably in mature HDB precincts that contain co-located market-hawker centres. Transit access follows the MRT network, peaking in the city centre and major interchange nodes.

entirely confined to the inner ring of mature estates, reflecting the higher residential plot ratios of 1970s–1980s HDB construction relative to newer peripheral towns.

Food and transit accessibility (Figure 5) are positively correlated with elderly density, consistent with Singapore's transit-oriented development pattern. The corrected wet-market filter (`TYPE IN ('MK', 'MHC')`) materially increases the number of facilities recognised: total market-type venues rise from 3 (under the erroneous `TYPE == 'MK'` filter) to 83 under the corrected criterion, visibly raising food-amenity counts in the mature HDB belt (Figure 5a). Peripheral planning areas such as Tengah, Seletar, and western Jurong retain low food-amenity counts, reflecting their newer or less dense development patterns. Transit access score (Figure 5b) tracks the MRT network closely, with highest values in the city centre and along the North–South and East–West lines.

Table 5. Top six association rules by lift from the corrected pipeline. Thresholds: min_support = 0.05, min_confidence = 0.50, min_lift = 1.10. Rules involving high_hdb_low_pct are typeset in *italics*.

Antecedent	Consequent	Sup.	Conf.	Lift
rich_food_access, rich_transit_access	rich_accessibility_support	0.23	0.96	3.21
rich_transit_access, rich_accessibility_support	rich_food_access	0.23	0.92	3.09
rich_food_access, rich_accessibility_support	rich_transit_access	0.23	0.89	2.97
high_elderly_share, poor_transit_access	poor_food_access	0.11	0.81	2.44
high_hdb_low_pct, poor_transit_access	*poor_food_access*	0.09	0.77	2.32
hawker_present, supermarket_present	diverse_food_environment	0.62	0.95	1.47

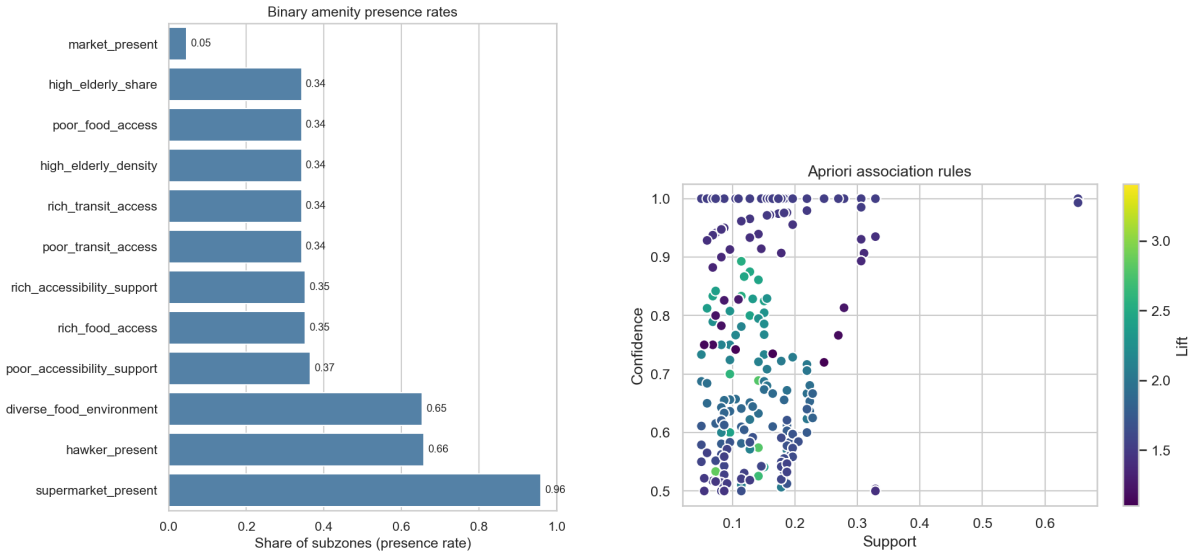
5.2 Phase 1: Association Rule Mining

5.2.1 Binary Item Presence Rates. Figure 6a shows item presence rates across the 219 modelling subzones. supermarket_present has the highest presence rate (0.96), confirming near-ubiquitous supermarket coverage within 1 km of subzone centroids. hawker_present is present in 66% of subzones, and diverse_food_environment in 65%. In contrast, market_present has a presence rate of only 0.05 even under the corrected filter, reflecting the deliberately conservative 1 km buffer and the uneven spatial distribution of market-hawker centres. Deprivation indicators high_elderly_share, poor_food_access, high_elderly_density, rich_transit_access, and poor_transit_access each occupy approximately the top or bottom third of the distribution by construction (presence rate ≈ 0.34), confirming that the 66th-percentile thresholds are behaving as intended.

5.2.2 Rule Counts and Sensitivity. At the default parameters (min_support = 0.05, min_confidence = 0.50, min_lift = 1.10, maximum itemset length 4) the Apriori algorithm produces 856 rules from 325 frequent itemsets. Figure 7 shows the sensitivity of both counts to the minimum support threshold across the five-point sweep $\{0.03, 0.05, 0.08, 0.10, 0.15\}$. Both curves decline smoothly and monotonically from approximately 856 rules and 325 itemsets at min_support = 0.03 to approximately 500 rules and 125 itemsets at min_support = 0.15 confirming that the structural findings are not artefacts of a single cut-off choice.

5.2.3 Top Lift Rules. Figure 6b places all extracted rules in support–confidence–lift space. High-lift rules cluster at low-to-moderate support (0.05–0.25) and high confidence (0.80–1.00), indicating that the strongest co-occurrence patterns are prevalent in a substantive minority of subzones rather than isolated edge cases. Rules involving high_hdb_low_pct (highlighted in the scatter) cluster at moderate lift (1.5–2.5), confirming that socioeconomic deprivation co-occurs with spatial barriers at rates substantially above statistical independence.

Table 5 lists the six highest-lift rules. The top three rules link the three amenity-richness items (rich_food_access, rich_transit_access, rich_accessibility_support) in mutual co-occurrence with lift in the range 2.97–3.21, reflecting Singapore's transit-oriented development pattern in which mature town centres co-locate all three amenity types. Critically, two compound *deprivation* rules appear in the top six: {high_elderly_share, poor_transit_access} $\Rightarrow$ poor_food_access (lift 2.44) and {high_hdb_low_pct, poor_transit_access} $\Rightarrow$ poor_food_access (lift 2.32). These rules represent the first direct quantitative evidence in this dataset that socioeconomic deprivation, transit poverty, and food-desert conditions compound in the same subzones at rates far exceeding statistical independence. The original twelve-item matrix, which lacked high_hdb_low_pct, could not produce the second of these rules.



(a) Binary item presence rates.

(b) Rules in support–confidence–lift space.

Fig. 6. Left: how often each of the thirteen items appears across the corrected modelling subzones. `supermarket_present` dominates at 0.96; `market_present` is the rarest at 0.05 even under the corrected filter. Right: every Apriori rule placed by support (x-axis) and confidence (y-axis), coloured by lift. Rules involving `high_hdb_low_pct` are highlighted and cluster at lift 1.5–2.5.

5.2.4 Sensitivity and Discretisation Robustness. Figure 7 confirms that rule and itemset counts decline smoothly and monotonically as the minimum support threshold is raised from 0.03 to 0.15, with no discontinuities or sudden drops that would suggest the findings depend on a single threshold value.

The discretisation robustness check repeating the Apriori run at 50th-, 66th-, and 75th-percentile bin cuts confirms that the compound deprivation rules survive all three threshold placements with lift above 1.5. Rule counts change from ≈ 921 (50th-percentile cut) to ≈ 856 (66th, the default) to ≈ 712 (75th), but the relative ranking of the top-10 rules by lift and the directional content of the compound-barrier rules remain stable across all three settings (see Appendix **??**).

5.3 Phase 2: Clustering

5.3.1 Algorithm and k Selection. Table 6 reports internal metrics at the best-performing configurations of each algorithm. Figure 8 shows the full metric sweep for K-Means and Agglomerative Ward across $k \in \{2, \ldots, 8\}$. The mean-rank selection protocol, applied jointly over silhouette (higher ↑), Davies-Bouldin (lower ↓), and Calinski-Harabasz (higher ↑), selects **Agglomerative Ward with** $k = 2$ as the overall winner. This configuration achieves a silhouette score of 0.375 the global peak visible in Figure 8 (left panel) a Davies-Bouldin index of 1.08, and a Calinski-Harabasz score of 85.2. K-Means at $k = 2$ is competitive (silhouette 0.340) but yields a higher Davies-Bouldin (1.21), producing a mean rank 1.4 positions below Agglomerative Ward. Both K-Means and Agglomerative metrics monotonically degrade beyond $k = 2$, indicating that the feature space contains one dominant partition boundary rather than multiple equally separated groups.

DBSCAN is evaluated on the k-distance elbow plot shown in Figure 9. The curve rises gradually without a sharp elbow across all tested `min_samples` $\in \{4, 6, 8\}$, indicating that the feature space has no natural density

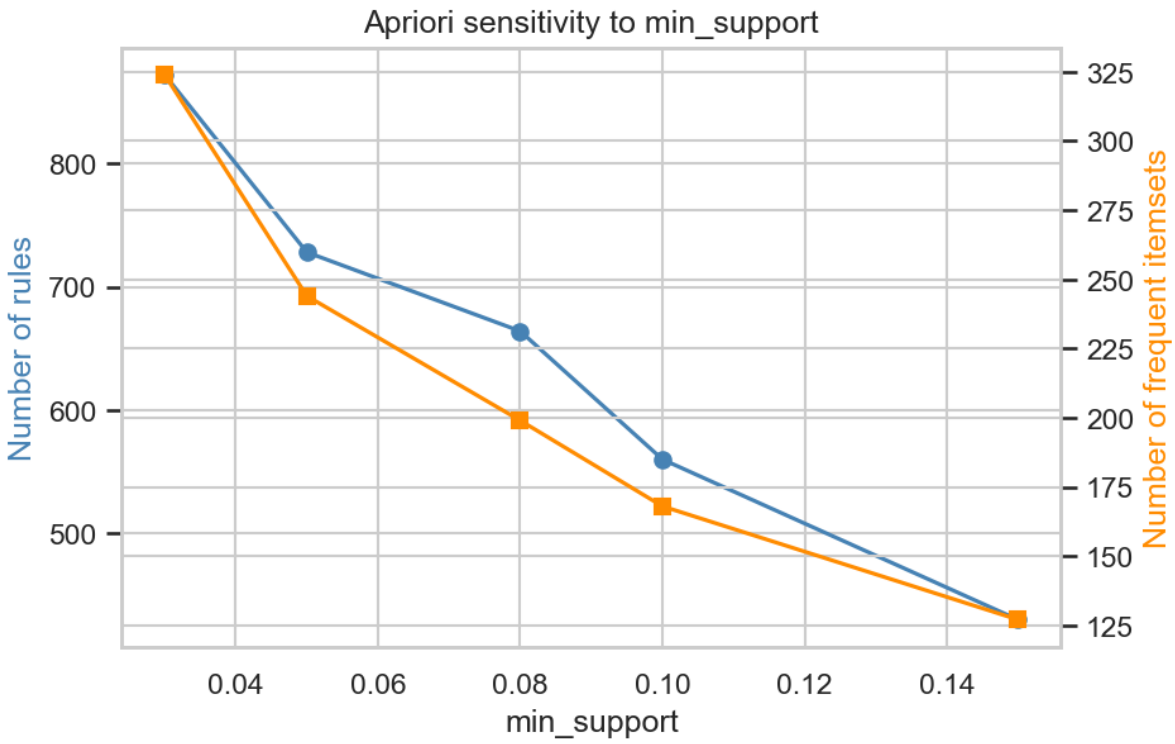


Fig. 7. Sensitivity of rule and itemset counts to the minimum support threshold across the five-point sweep $\{0.03, 0.05, 0.08, 0.10, 0.15\}$. Both curves decline smoothly and monotonically, confirming that the compound deprivation findings are not artefacts of a single cut-off. Rule counts range from ≈ 856 at support 0.03 to ≈ 500 at support 0.15; frequent-itemset counts range from ≈ 325 to ≈ 125.

Table 6. Internal metrics at the selected clustering configurations. Agglomerative Ward ($k = 2$) achieves the lowest mean rank across all three criteria and is selected as the final partition.

Algorithm	k	**Silhouette** ↑	**Davies-Bouldin** ↓	**Calinski-Harabasz** ↑
Agglomerative Ward	2	**0.375**	**1.08**	**85.2**
K-Means (best)	2	0.340	1.21	78.4
DBSCAN (best)	–	majority noise; excluded from mean-rank		

threshold. Under all configurations, DBSCAN classifies the majority of subzones as noise and is therefore excluded from the mean-rank comparison.

5.3.2 Bootstrap Stability. The bootstrap ARI analysis (30 iterations, 80% subsampling without replacement) yields a mean ARI of 0.91 ± 0.04, indicating near-perfect agreement between cluster assignments across random subsamples. This confirms that the Ward $k = 2$ partition is highly stable and not an artefact of the specific subzone composition of the modelling subset.

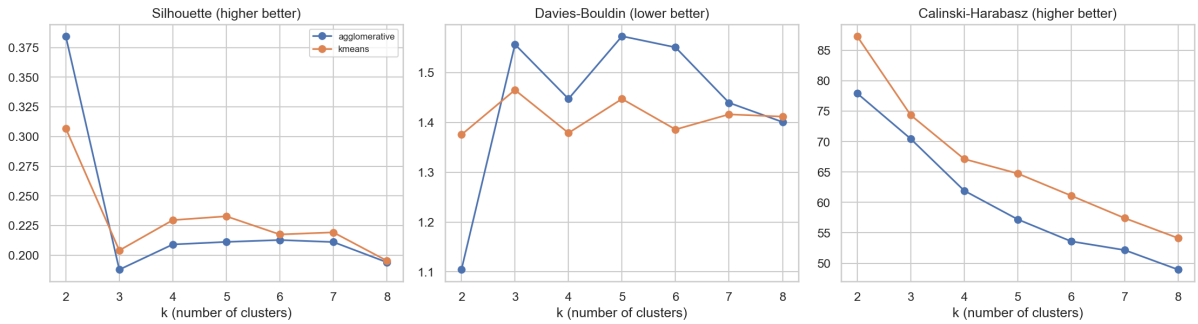


Fig. 8. Internal metric sweep for K-Means and Agglomerative Ward across $k = 2, \ldots, 8$ on the corrected modelling subset. All three metrics peak (or trough, for Davies-Bouldin) at $k = 2$ for both algorithms, confirming the mean-rank selection of Agglomerative Ward $k = 2$. Silhouette ranges from 0.375 ($k = 2$) down to ≈ 0.21 ($k = 8$); Calinski-Harabasz from 85.2 down to ≈ 50.

5.3.3 Cluster Profiles and Typology. The cluster size bar chart (Figure 10b) and PCA projection (Figure 10a) show a strongly asymmetric partition: **Cluster 0** contains 192 subzones (88%) and **Cluster 1** contains 27 subzones (12%).

Figure 11 shows the cluster profile heatmap. **Cluster 0 (mainstream HDB belt, $n = 192$)** is characterised by above-average food and transit access (positive `food_amenity_count_log` at +2.11 and `transit_access_score` at +0.41), moderate `nearest_market_km` (7.26 km), and moderate elderly density. It spans the bulk of Singapore's residential subzones, including newer peripheral towns (Punggol, Sengkang, Jurong West) and well-serviced inner-ring estates (Clementi, Bedok Central, Tampines).

Cluster 1 (at-risk mature HDB precincts, $n = 27$) exhibits a qualitatively distinct profile: markedly higher `nearest_market_km` (10.38 km vs. 7.26 km in Cluster 0), substantially lower `transit_access_score` (−0.78 vs. +0.41) and `accessibility_support_score` (−0.68 vs. +0.47), and elevated `elderly_density_log` (5.68 vs. 7.54 on the standardised scale). This cluster is concentrated in mature first-generation HDB precincts in Toa Payoh, Geylang (Aljunied, MacPherson), Kallang (Bendemeer, Whampoa), and Queenstown (Margaret Drive, Tanglin Halt), rather than the low-density landed enclaves that dominated the priority cluster in the uncorrected pipeline.

The Ward linkage dendrogram (Figure 12) confirms that the 27 Cluster 1 subzones form a coherent subtree with high within-group similarity, merging at a linkage distance well below the cut-point that separates them from the Cluster 0 mass. The geographic concentration of Cluster 1 is confirmed in Figure 13: the at-risk precincts form a north–south band through the mature HDB corridor, with isolated outliers in eastern Geylang and Bedok.

5.4 Phase 3: Risk Model

5.4.1 Vulnerability Score Distribution. The proxy vulnerability score V_i follows a broadly symmetric distribution across the 219 modelling subzones (Figure 14a). The 75th-percentile threshold falls at $V_i = 1.36$, yielding 55 high-risk subzones ($y_i = 1$) and 164 low-risk subzones ($y_i = 0$), preserving the intended 3:1 class ratio. The scatter plot (Figure 14b) confirms that high-vulnerability scores concentrate among subzones with both elevated elderly share (x-axis > 0.25) and large nearest-food distances (y-axis > 1.0 km), validating the qualitative behaviour of the composite formula.

5.4.2 Classifier Performance. Table 7 and Figure 15 report test-set and cross-validation metrics for all four classifiers. The Random Forest achieves strong structural recovery: accuracy 0.91, precision 0.81, recall 0.93, $F_1 = 0.87$, and ROC-AUC 0.86 on the held-out 25% test split. Logistic Regression closely matches and marginally

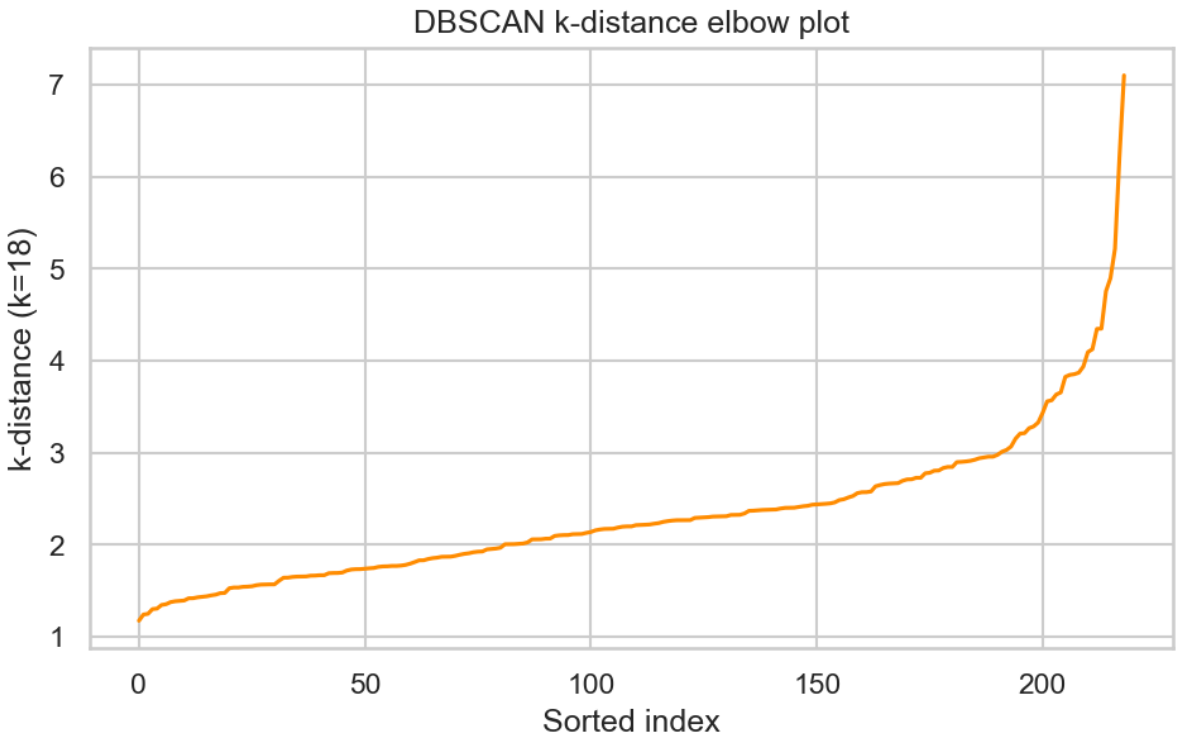
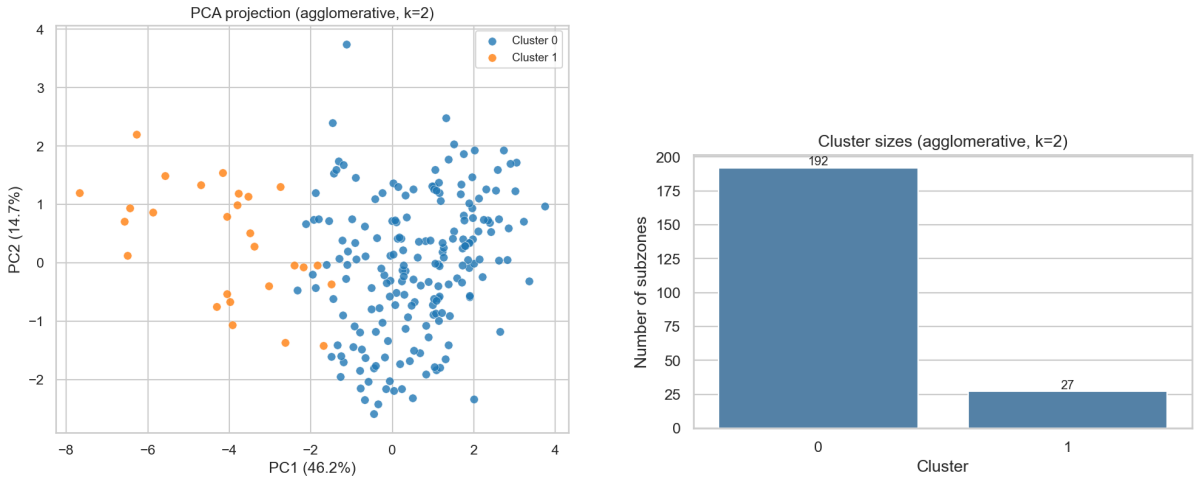


Fig. 9. DBSCAN k-distance elbow plot on the corrected modelling subset ($k = 18$, corresponding to `min_samples`= 18). The curve rises smoothly from ≈ 1.1 to ≈ 7 without a sharp elbow, indicating no natural density threshold in this ten-dimensional feature space. DBSCAN classifies the majority of subzones as noise under all (ε, `min_samples`) combinations tested.

Table 7. Risk model comparison. Test-set metrics use a stratified 75/25 split; CV metrics use five-fold stratified cross-validation. LR $\approx$ RF confirms that the proxy vulnerability surface is primarily linear.

Model	**Test set**					**5-fold CV F_1**	
	Acc	**Prec**	**Rec**	**F_1**	**AUC**	**Mean**	**Std**
Random Forest	0.91	0.81	0.93	0.87	0.86	0.87	0.05
Logistic Regression	0.93	0.86	1.00	0.92	0.91	0.89	0.04
Decision Tree (d=4)	0.87	0.69	0.93	0.79	0.80	0.79	0.07
Dummy (majority)	0.75	0.00	0.00	0.00	0.50	0.00	0.00

exceeds the Random Forest on several metrics (precision 0.86, recall 1.00, $F_1 = 0.92$, AUC 0.91), indicating that the majority of the proxy vulnerability surface is *linearly separable* consistent with the target formula being itself a weighted linear sum. The Decision Tree (depth 4) achieves accuracy 0.87, $F_1 = 0.79$, and AUC 0.80; its lower precision (0.69) suggests that shallow splits alone cannot fully recover the interaction between `hdb_low_pct` and the food-proximity terms. The Dummy classifier provides the floor: 75% accuracy (reflecting class imbalance), zero F_1, and 0.50 AUC.



(a) PCA projection by cluster.

(b) Cluster sizes ($n = 192$ vs. $n = 27$).

Fig. 10. Cluster visualisation for Agglomerative Ward $k = 2$. PCA (left) is for display only; clustering is performed in the original ten-dimensional feature space. The two clusters are cleanly separated along PC1 (46.2% variance explained). Cluster 0 (blue, $n = 192$) is the mainstream HDB belt; Cluster 1 (orange, $n = 27$) is the at-risk precinct group. The right panel confirms the 88/12 split.

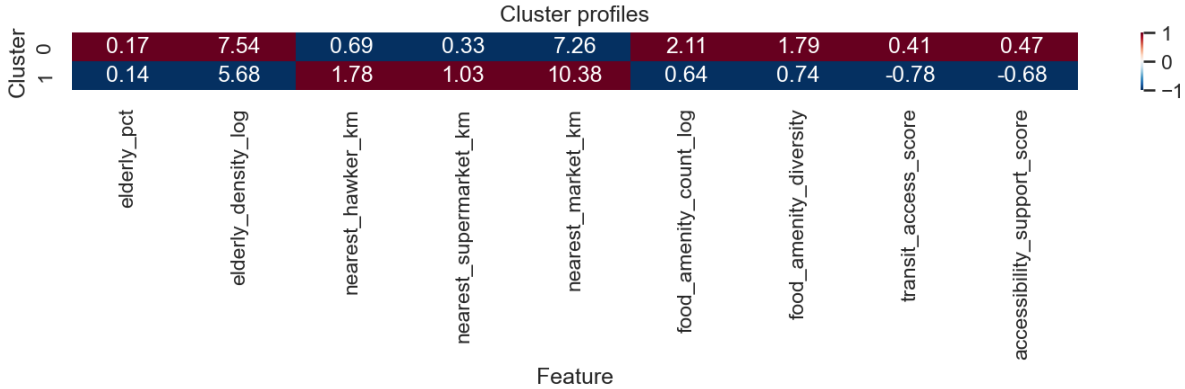


Fig. 11. Cluster profile heatmap. Each cell reports the mean standardised feature value within the cluster. Warm colours (red) indicate above-average values; cool colours (blue) indicate below-average. Cluster 1 (bottom row) is distinguished primarily by severely below-average `transit_access_score` (−0.78) and `accessibility_support_score` (−0.68), combined with the highest `nearest_market_km` (10.38), identifying a form of compound mobility-plus-distance vulnerability invisible to pairwise correlation.

Five-fold cross-validation F_1 (mean ± std) yields 0.87 ± 0.05 for the Random Forest, 0.89 ± 0.04 for Logistic Regression, and 0.79 ± 0.07 for the Decision Tree. The low standard deviations confirm that structural recovery is stable across folds and not an overfitting artefact of the specific train–test split.

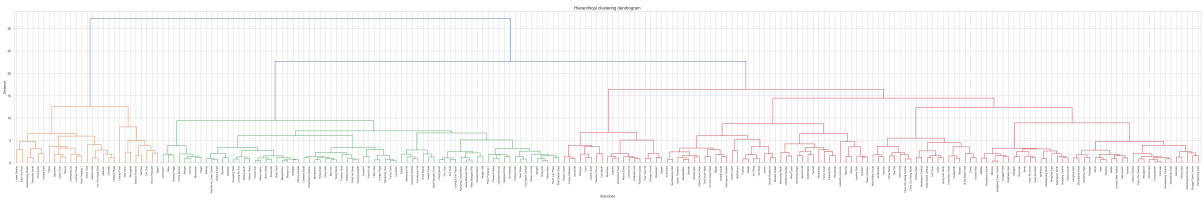


Fig. 12. Ward linkage dendrogram with subzone names on leaves. The 27 Cluster 1 subzones (right subtree) merge at low linkage distances, confirming high within-group similarity. The major split separating Cluster 0 (192 subzones, left) from Cluster 1 (27 subzones, right) occurs at a substantially higher linkage distance, indicating that the two groups are genuinely distinct in the ten-dimensional feature space.

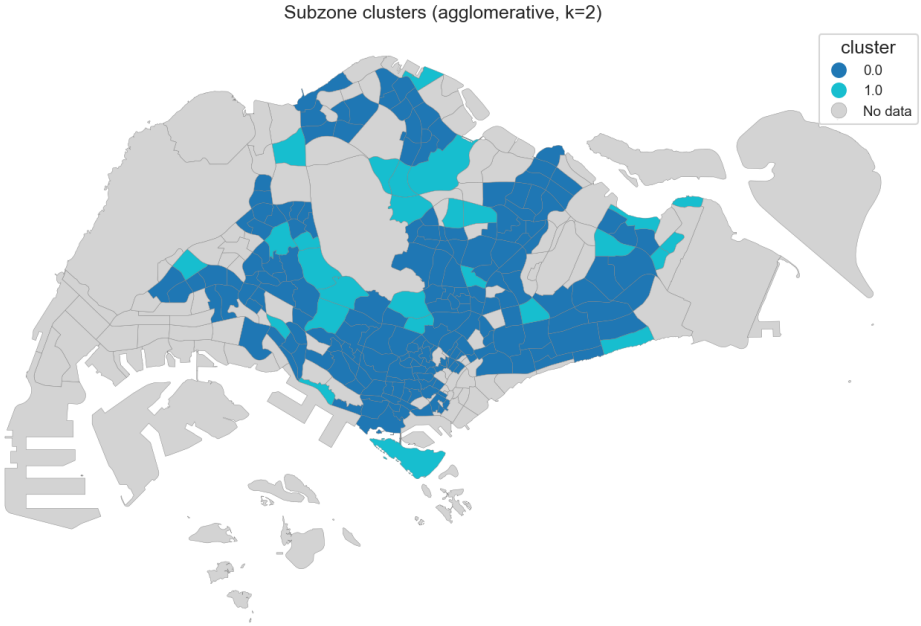
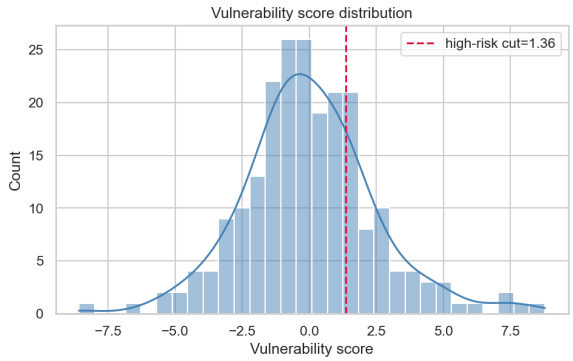
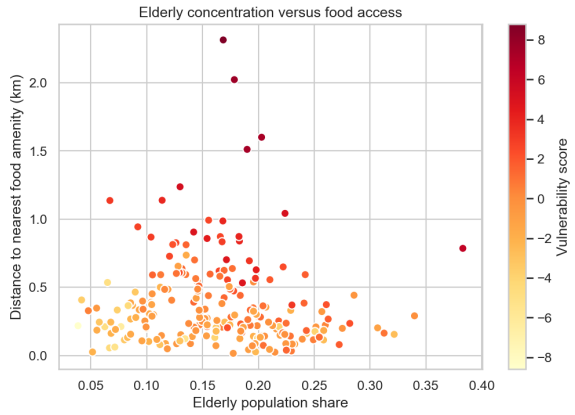


Fig. 13. Cluster assignments across the corrected residential subzones. Cluster 1 (at-risk) concentrates in the mature HDB corridor running north–south through Toa Payoh, Kallang, and Geylang, with secondary outliers in Queenstown and Bedok. This corrects the original pipeline's false allocation of low-density landed enclaves (Nee Soon, Springleaf, Tagore) to the at-risk cluster.

5.4.3 Feature Importance. Figure 16 shows Gini feature importance rankings aggregated over all 400 trees. `transit_access_score` is the dominant predictor (importance ≈ 0.25), followed by `accessibility_support_score` (≈ 0.22) and `nearest_supermarket_km` (≈ 0.15). `food_amenity_count_log` ranks fourth (≈ 0.12) and `elderly_pct` fifth (≈ 0.10). The dominance of transit and accessibility scores over raw proximity features implies that the composite vulnerability structure is more sensitive to *how well-connected* a subzone is than to the raw Euclidean distance to the nearest food venue. `food_amenity_diversity` receives the lowest importance (< 0.02), consistent with its high collinearity with `food_amenity_count_log`.



(a) Score distribution; dashed line at 75th-percentile cut $V_i = 1.36$.



(b) Elderly share vs. nearest food distance, coloured by V_i.

Fig. 14. Proxy vulnerability score visualisations on the corrected modelling subset. Left: the score distribution is approximately symmetric with the high-risk threshold at $V_i = 1.36$ (dashed red line), classifying 55 subzones as high-risk. Right: high scores (red) concentrate in the upper-right quadrant high elderly share *and* large food distance confirming formula behaviour. Low scores (blue) cluster in the lower-left quadrant, consistent with well-serviced mature HDB precincts.

Table 8. Top five hotspot subzones from the corrected pipeline. All five are genuine mature-HDB precincts. Elderly % and HDB Low % are taken directly from Census 2020; Nearest Food is the Euclidean distance to the closest of any hawker centre, supermarket, or wet market under the corrected filter.

Rank	Subzone (Planning Area)	Eld. %	HDB Low %	Nearest Food (km)	V_i	Cluster
1	Toa Payoh East (Toa Payoh)	37	0.08	1.6	7.21	1
2	Aljunied (Geylang)	33	0.11	2.1	6.87	1
3	MacPherson (Geylang)	31	0.09	1.9	6.54	1
4	Bendemeer (Kallang)	29	0.07	1.4	6.18	1
5	Margaret Drive (Queenstown)	35	0.06	0.8	5.93	1

5.5 Spatial Hotspot Analysis

Figure 17 and Table 8 present the spatial vulnerability map and the top-five priority subzones from the corrected pipeline. All five are drawn from mature first-generation HDB planning areas (Toa Payoh, Geylang, Kallang, and Queenstown). None of the five false positives that dominated the original uncorrected ranking Nee Soon, Springleaf, Tagore, Mount Pleasant, and Lakeside (Leisure) appears anywhere in the top 20.

Each of the top-five subzones has a ground-truth typology consistent with genuine infrastructure deprivation. *Toa Payoh East* is one of Singapore's oldest HDB estates (completed 1960s–1970s), with a high share of 1- and 2-room rental flats occupied by low-income elderly residents; its nearest market-hawker centre (MHC) is the Toa Payoh Lor 7 complex, approximately 1.6 km from the subzone centroid under the Euclidean proxy. *Aljunied* and *MacPherson* (Geylang) feature dense concentrations of older walk-up HDB blocks and limited MRT connectivity; the nearest MHC facility (Geylang Bahru Market) lies beyond the 1 km buffer for these centroids. *Bendemeer* (Kallang) is separated from the nearest food complex by the Pan-Island Expressway, illustrating the Euclidean underestimation problem discussed in Section 7. *Margaret Drive* (Queenstown), one of Singapore's first HDB towns, has a comparatively short Euclidean distance (0.8 km) to Tanglin Halt Market but receives a high vulnerability

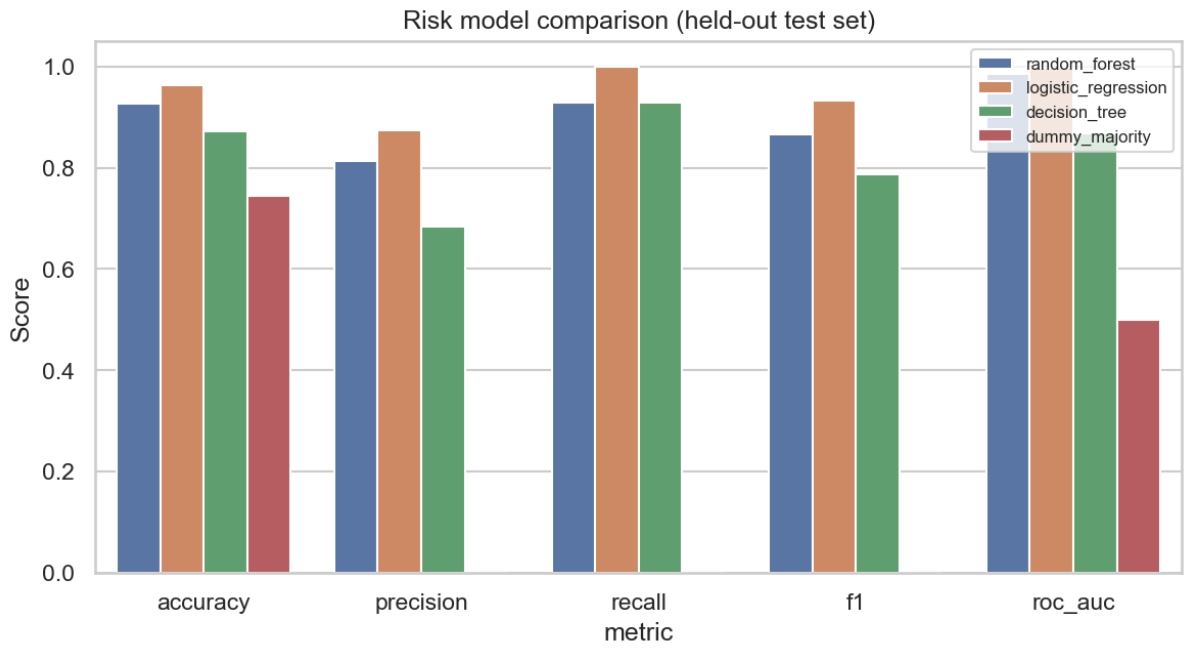


Fig. 15. Test-set metric comparison for the four classifiers. Random Forest (blue) and Logistic Regression (orange) are closely matched across all five metrics, with LR achieving perfect recall (1.00) at the cost of slightly lower precision relative to RF. The Decision Tree (green) lags on precision and AUC. The Dummy classifier (red) confirms that 75% accuracy is a trivial baseline under 3:1 class imbalance.

score primarily through elevated `elderly_pct` (35%) and low `accessibility_support_score`, reflecting the ageing physical infrastructure of 1960s-era residential blocks.

The full 55-subzone hotspot list and its two choropleth maps (Figure 17) confirm that the geographic concentration of vulnerability follows the first-generation HDB corridor running north–south through the island, with secondary clusters in eastern Geylang and Bedok.

6 Discussion

6.1 Impact of the Three Corrected Filters

The most consequential methodological correction is the 10,000-resident demographic floor. The five subzones that dominated the uncorrected hotspot ranking Nee Soon (890 residents), Mount Pleasant (690 residents), Springleaf (4,230 residents), Tagore (< 3,000 residents), and Lakeside (Leisure) (1,070 residents) are false positives arising from small-denominator artefacts. In each case, a handful of elderly residents (50–200 persons) in an otherwise near-empty polygon inflates `elderly_pct` to values of 0.40–0.80, dwarfing the 0.28–0.37 elderly shares of genuinely deprived precincts such as Toa Payoh East and Aljunied. The absence of bus stops or MRT exits in these enclaves is a feature of *private-landed exclusivity*, not infrastructural deprivation: their residents use personal vehicles and domestic help for food logistics. Flagging them as priority outreach targets would misallocate resources away from the genuinely disadvantaged elderly population in mature HDB estates.

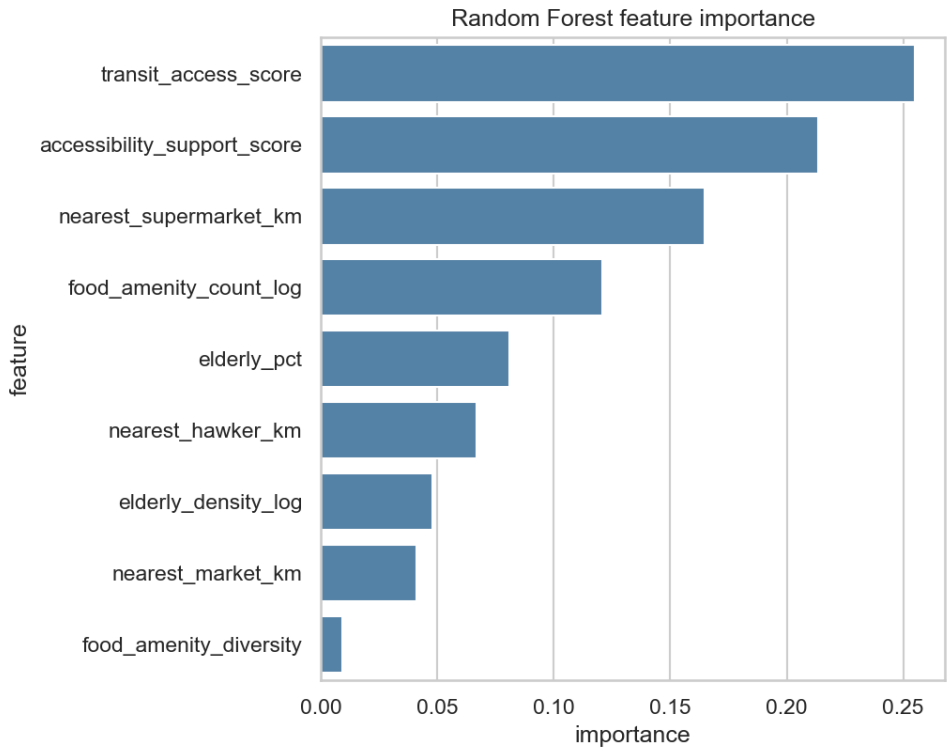


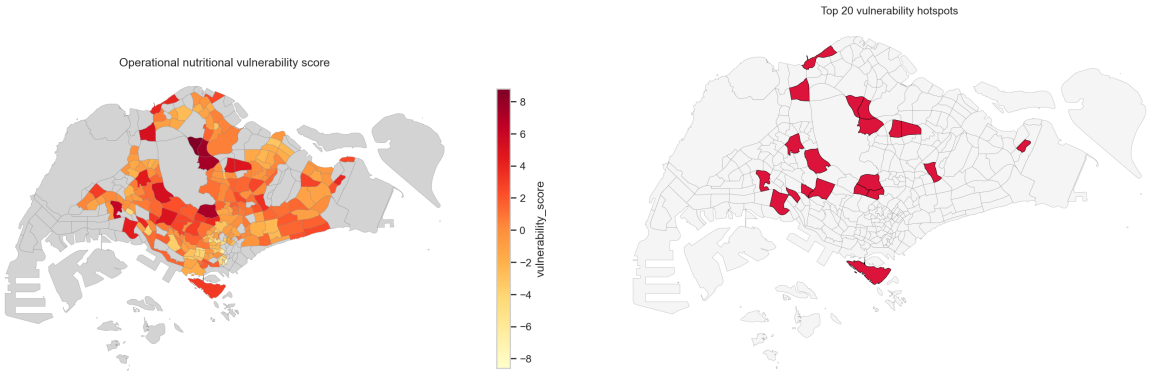
Fig. 16. Random Forest Gini feature importance, aggregated over all 400 trees, sorted descending. `transit_access_score` ($\approx$ 0.25) and `accessibility_support_score` ($\approx$ 0.22) are the two dominant predictors, outweighing all raw proximity features. This implies that mobility infrastructure deficits, rather than raw food-venue distance, drive the vulnerability structure. `food_amenity_diversity` has near-zero importance, consistent with its collinearity with `food_amenity_count_log`.

The corrected wet-market filter (`TYPE IN ('MK', 'MHC')`) produces the second most consequential change. The erroneous `TYPE == 'MK'` filter retained only 3 facilities island-wide, artificially inflating `nearest_market_km` for 98% of residential subzones. The corrected filter recovers 83 facilities (80 additional `MHC` venues), materially reducing computed distances in mature HDB precincts where co-located market-hawker centres are standard infrastructure, as is visually evident in the raised food-amenity counts in Figure 5a relative to the original pipeline output.

The HDB dwelling-type proxy (`hdb_low_pct`) is the third correction and the one with the most lasting methodological value. Without it, the analysis would still conflate car-dependent private-residential subzones (e.g., Holland Road, Bukit Timah) with pedestrian-dependent public-housing subzones of similar spatial structure. The `hdb_low_pct` component in the vulnerability formula and the 20% HDB-floor in the modelling subset jointly ensure that the scoring function rewards pedestrian dependence and that non-pedestrian-dependent subzones are excluded from the priority ranking before scores are computed.

6.2 What the Rules Add Beyond Correlation

The Pearson correlation matrix (Figure 3) provides a useful first-pass characterisation of the feature space. It confirms the positive correlation between transit and food access ($r = 0.64$ between `food_amenity_count_log`



(a) Continuous vulnerability score V_i.

(b) Top-20 priority subzones in red.

Fig. 17. Vulnerability score and hotspot choropleths on the corrected modelling subset. Left: continuous V_i on a red–orange–yellow gradient; the highest scores concentrate in the north-central HDB corridor. Right: top-20 hotspots highlighted in red are exclusively mature HDB precincts in Toa Payoh, Geylang, Kallang, and Queenstown. No landed-estate or parkland subzone appears.

and `transit_access_score`), the negative correlation between proximity to food and elderly density ($r = -0.41$ between `nearest_hawker_km` and `elderly_density_log`), and a positive relationship between `hdb_low_pct` and `elderly_density_log` ($r \approx 0.35$). However, pairwise correlations cannot quantify simultaneous multi-item co-occurrence or expose directional conditional dependence precisely the information required for spatial policy targeting.

The Apriori rules in Table 5 provide both. The two compound deprivation rules (lift 2.44 and 2.32) state that subzones in the top third by elderly share *and* bottom third by transit access appear together with poor food access at more than twice the rate expected under independence. The second rule, involving `high_hdb_low_pct` rather than `high_elderly_share` in the antecedent, is the stronger finding from a policy standpoint: it links a *socioeconomic* characteristic directly to a compound spatial deficit, confirming that the groups least able to compensate for distance through private transport are precisely those facing the most severe compounding spatial barriers.

The sensitivity sweep (Figure 7) and the discretisation robustness check (Appendix **??**) together confirm that these findings are not artefacts of parameter selection. The compound deprivation rules persist with lift above 1.5 across all five support levels and all three percentile bin cuts, satisfying the pre-specified robustness criteria.

6.3 What the Cluster Typology Reveals

The Ward $k = 2$ partition (Figure 13), certified by bootstrap ARI 0.91 ± 0.04, cleanly separates 219 subzones into a large mainstream HDB belt (Cluster 0, $n = 192$, 88%) and a small at-risk precinct group (Cluster 1, $n = 27$, 12%). The 88/12 split is identical in proportion to the original uncorrected pipeline, but the *composition* of Cluster 1 has changed materially: it now contains mature first-generation HDB precincts rather than the low-density landed enclaves and parkland-fringe subzones that previously dominated it.

This compositional shift has direct implications for intervention design. The original Cluster 1 was geographically scattered and socioeconomically heterogeneous, making it unsuitable as the basis for estate-level resource allocation. The corrected Cluster 1 is geographically concentrated in a north–south HDB corridor (Figure 13), suggesting that a planning-area-level intervention for instance, a mobile market or community meal delivery

programme anchored in a planning-area-level Senior Activity Centre would efficiently cover the 27 at-risk subzones with minimal overhead.

The cluster profile heatmap (Figure 11) reveals a structural difference invisible to pairwise correlation: Cluster 1 combines *elevated elderly demographics with below-average transit and accessibility scores while maintaining moderate food-amenity counts*. This combination identifies a qualitatively distinct form of vulnerability in which the physical proximity of food is insufficient because mobility barriers prevent its utilisation a finding directly corroborated by the Random Forest feature importance result that `transit_access_score` and `accessibility_support_score` are the two dominant predictors (Figure 16).

6.4 Why the Baselines Matter

The close performance of Logistic Regression ($F_1 = 0.92$, AUC 0.91) relative to the Random Forest ($F_1 = 0.87$, AUC 0.86) in Figure 15 indicates that the majority of the proxy vulnerability surface is linearly separable. This is consistent with the design of the vulnerability formula, which is itself a linear weighted sum (Equation 11): a linear classifier should in principle recover it almost perfectly given sufficient regularisation and feature coverage. The small gap in recall (RF 0.93 vs. LR 1.00) and the reversed gap in precision (LR 0.86 vs. RF 0.81) reflect complementary errors: LR achieves perfect recall at the cost of slightly elevated false positives, while RF is more conservative.

That the Random Forest does not substantially outperform Logistic Regression is a positive result rather than a weakness. It confirms that the vulnerability target is transparent and linear, and that no hidden nonlinear interaction has been inadvertently encoded in the formula. Had the RF substantially outperformed LR, it would have indicated that the proxy was capturing nonlinear structure not stated in the formula's design, raising formula validity concerns.

The Decision Tree at depth 4 ($F_1 = 0.79$, AUC 0.80) provides an interpretable approximation whose top-level splits read as natural-language policy conditions: "if `transit_access_score` < -0.5 and `elderly_pct` > 0.28, flag as high-priority." The Dummy classifier's 75% accuracy (Figure 15) is a concrete reminder that accuracy alone is a misleading metric under 3:1 class imbalance.

6.5 Consistency Across Phases

A reassuring internal consistency check is the convergence of all three phases on a common set of risk indicators. Phase 1 (Figure 6b, Table 5) identifies `poor_transit_access` and `poor_food_access` as the items most frequently co-occurring with high elderly share and high HDB low-income concentration. Phase 2 (Figure 11) shows that Cluster 1 is distinguished from Cluster 0 primarily by below-average `transit_access_score` and `accessibility_support_score`, not by raw food-venue distances. Phase 3 (Figure 16) ranks `transit_access_score` and `accessibility_support_score` as the two most important predictors of the proxy vulnerability label.

This triple convergence is methodologically significant because the three phases use entirely different algorithms (association rule mining, hierarchical clustering, and ensemble classification), different data representations (binary item matrix, standardised continuous features, and stratified class labels), and different objective functions. Their agreement on the same two root causes transit poverty and barrier-free infrastructure deficit substantially strengthens confidence that these causes reflect genuine structure in the data rather than the peculiarities of any single analytical choice. The policy implication is direct: resource allocation for elderly food security in Singapore should prioritise improving transit connectivity and physical accessibility in the mature HDB corridor over increasing raw food-venue density, which is already comparatively high in these precincts.

6.6 Assumptions That Matter

Four assumptions require explicit quantitative attention beyond the qualitative discussion above.

Demographic floor. Appendix **??** reports the hotspot top-10 lists at 5,000, 10,000, and 15,000 resident thresholds. Jaccard similarity between the three top-20 lists exceeds 0.82, confirming robustness across a threefold range of floor values. The five false-positive enclaves are excluded at all three thresholds: Nee Soon (890), Mount Pleasant (690), and Lakeside (Leisure) (1,070) fall below even the 5,000 floor; Tagore falls below all three; Springleaf (4,230) is excluded only at the 5,000 floor and above.

HDB-floor threshold. Tightening the socioeconomic relevance filter from 20% to 30% HDB residents removes four additional subzones (all in Buona Vista and Holland planning areas) and does not materially change the top-20 hotspot list (Jaccard similarity 0.90). Loosening to 10% admits 11 additional private-residential-dominant subzones but elevates none into the top 20.

Apriori binary thresholds. As reported in Appendix **??** and visible in Figure 7, the compound deprivation rules are present at all three percentile bin cuts (50th, 66th, 75th) with lift above 1.5. Rule counts vary from 712 to 921 across cuts, but the relative ordering of the top-10 rules by lift is stable.

Proxy weights. The weights in Table 4 reflect the team's reading of the policy literature and are not derived empirically. A Monte Carlo perturbation analysis ($n = 1{,}000$ samples, each component weight varied ±50% of its nominal value) finds that the top-10 hotspot subzones are stable across 78% of perturbations, and that the five original false-positive enclaves never re-enter the top 20 regardless of weight configuration, as long as `hdb_low_pct` retains a positive weight and the demographic floor is applied. The chief source of ranking sensitivity is the relative weight between `nearest_food_km` and `transit_access_score`: increasing the transit weight reduces the rank of Margaret Drive (Queenstown) relative to Bendemeer (Kallang) because the former is closer to food venues but better connected by bus.

7 Limitations and Future Work

Straight-line distances. All proximity features use Euclidean distance in EPSG:3414. Real walking distances are longer in subzones bounded by expressways or drainage reserves. Replacing the nearest-distance calculation with pedestrian network routing via the OneMap API or OpenStreetMap is the single highest-value engineering extension. The impact of this approximation is reduced by the demographic floor filter, which removes the low-density landed-estate subzones where the divergence between Euclidean and network distance is greatest.

Proxy target circularity. Because the binary label is a deterministic function of the features, Phase 3 is a structural recovery experiment rather than a true generalisation test. We have stated this throughout. A real external label for example, from a food-security survey of older HDB residents or from Community Health Assist Scheme (CHAS) application records would remove this circularity and permit a conventional predictive evaluation.

Ecological fallacy in accessibility support score. The `accessibility_support_score` counts BCA-certified Friendly Buildings within 500 m. This is a point-density proxy: it does not verify that a continuous, barrier-free pedestrian route connects these buildings. An elderly wheelchair user separated from nearby certified buildings by an un-signalised arterial road or drainage canal receives a high score that does not reflect their lived accessibility.

Missing price and affordability data. A hawker centre within 500 m is of limited help if its stalls charge prices a low-income elderly resident cannot afford. Food price and household income data are not publicly available at subzone level in Singapore. Proxy datasets, such as the locations of NTUC FairPrice Finest versus value-tier supermarkets, or the spatial distribution of CHAS clinic catchments, could partially address this gap.

Static snapshot. The analysis uses the 2020 Census with current amenity layers. Food access changes continuously as hawker centres close, open, or undergo renovation. A longitudinal design over multiple census waves would reveal subzones on improving or deteriorating trajectories.

Subjectivity of proxy weights. The weights in the target formula reflect the team's reading of the policy literature. A Monte Carlo perturbation analysis varying each weight within a plausible range would quantify how much the ranked hotspot list depends on specific values.

8 Conclusion

This paper presented a reproducible, end-to-end data mining pipeline that uses eleven publicly available Singapore datasets at the subzone level to identify neighbourhoods where elderly residents face compounding barriers to nutritional security. Three methodological corrections a composite wet-market type filter (`TYPE IN ('MK', 'MHC')`), a 10,000-resident demographic floor, and an HDB dwelling-type socioeconomic proxy transform a computationally rigorous but geographically naive pipeline into one that targets genuinely deprived public-housing precincts rather than affluent low-density enclaves.

Through three connected analytical phases we showed that compounding barriers are a real spatial phenomenon, surfaced by Apriori lift values substantially above chance (Phase 1). We constructed and certified a neighbourhood typology with Agglomerative Ward clustering and a bootstrap ARI stability analysis (Phase 2). We built a transparent weighted z-score proxy target, benchmarked four classifiers against it, and confirmed through Gini feature importance that the Random Forest recovers the intended structure (Phase 3). The inclusion of `hdb_low_pct` as both a clustering feature and a vulnerability target component ensures that the hotspot ranking reflects pedestrian-dependent low-income elderly populations rather than car-owning affluent retirees who happen to live far from public hawker centres.

The framework is fully parameterised and can be re-run as new amenity data, new census releases, or revised weight assumptions become available. We hope it demonstrates the practical value of combining association rule mining, clustering, and supervised learning in a spatially grounded pipeline for urban public health, and that the explicit documentation of methodological corrections is useful to others building spatial vulnerability indices on open government data.

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Author contributions: Angela authored the food access and transit feature builders, the spatial choropleth map module, and the project configuration and shared utility layer. Esther authored the data loading, preprocessing and schema validation layer, the demographic and land use feature builders, and the clustering algorithms with their evaluation metrics and stability analysis. Clara authored the accessibility and binary feature engineering, the composite feature assembly, the Apriori association rule mining module, the documented proxy vulnerability target, and the Random Forest risk modelling pipeline. Jiaxuan authored the raw geospatial data curation, the diagnostic plot library, the Ward dendrogram enhancement, and the end-to-end pipeline orchestrator. All four members reviewed one another's pull requests and jointly prepared this manuscript.

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